

Elasticsearch

A distributed
multitenant full-text
search engine

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Any idea how many products Amazon currently lists for sale? About a second ago they would have roughly 562 million products, and that number changes by the second! Well how then, do you get your favourite fish food when you search for it, less than within a second?

You say magic, I say search. While traditionally programmers have struggled with data volumes when implementing search in their applications (yes, SQL devs I am looking at you!), recent changes have made significant advances in altering the way we optimise and search through millions and even billions of documents in a matter of seconds.



One such leading technology is Elasticsearch. It is an enterprise-grade and readily scalable search engine capable of running very complex queries in an extremely fast response time.

Queries that would take 100s of seconds on a traditional SQL platform to run would take a few milliseconds on Elasticsearch (no, seriously!). To understand how this miracle happens, before we try it for ourselves, we have to understand the key principle behind Elasticsearch speed – Indexes.

Elasticsearch indexes each document in its repository. During an indexing operation, Elasticsearch

converts raw data such as log files or message files into internal documents and stores them in a basic data structure. Each document is a simple set of correlating keys and values: the keys are strings, and the values are one of numerous data types—strings, numbers, dates, or lists.

While the actual implementation is quite complex, at its heart a full text search index consists of one or more inverted indices. When a particular string of text is indexed, it is divided into tokens, and each token becomes an entry in the lookup table for the index. A token is associated with pointers to the documents in which that token appears. How the text is divided into tokens depends on the analyser used.

DISTRIBUTED ARCHITECTURE AND SCALING

Elasticsearch scales up to thousands of servers and accommodates petabytes of data. Its enormous capacity results directly from its elaborate, distributed architecture. And yet the ES user remains unaware of nearly all of the

*CODING MATTERS



* Better face detection in Amazon Rekognition

Updates to Amazon's face detection software have led to a significant increase in its performance in detecting faces from images.

<https://digit.in/RekogUpdt>



*Pyro now on Watson Machine Learning

Pyro, a universal probabilistic programming language (PPL) written in Python, is on IBM's Watson Machine Learning platform with PyTorch 0.4.

<https://digit.in/PyroWatson>



* GitHub improves merge conflict resolution

More automated merge conflict resolution has been added to the GitHub Desktop client, along with an easier way to create new repositories and merges.

<https://digit.in/GitDesk>